HR PEOPLE ANALYTICS · DATA STORYTELLING

The 100,000-Person Question

What a decade of employee records reveals about pay, performance and attrition — and the HR myths the data dismantles.

Presented by

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100,000 EMPLOYEES 20 columns · 100,000 rows	**9** DEPARTMENTS 7 job titles	**2014 → 2024** SNAPSHOT WINDOW Sep 2014 → Sep 2024 (partial years)
24 SQL QUERIES 7 sections · 32 output CSVs	**6** DATA-QUALITY CHECKS grain · validity · reconciliation	**21** CHARTS + DASHBOARDS Power BI · 4-tab dashboard

What changed in this edition

A complete SQL analytics layer (24 queries, 6 data-quality checks, 32 committed result files), a reproducible modelling chapter, and corrections to three published claims: attrition rates that were divided by the wrong denominator, a promotion rate stuck at 100%, and a “31% hiring collapse in 2024” that was a partial-year artefact — the real change is +2.4% versus 2023. Chapter 5 documents every fix; chapter 6 shows what the models do and do not explain.

Contents

Executive Summary	The five things to remember
1 · Workforce & Hiring	Headcount, cohorts and the missing hiring cliff
2 · The Compensation Puzzle	Education, gender, tenure and role
3 · The Performance Machine	Ratings, promotions, training ROI
4 · The Attrition Enigma	Where turnover hides — and doesn't
5 · The SQL Analytics Layer	Schema, 24 queries, QA suite, corrections
6 · What the Models Say	Can we predict churn? What sets pay?
7 · Myths vs. Math	Belief scoreboard
8 · What I Would Tell the CEO	Eight recommendations
9 · Method, Data Quality & Limits	How this was built — and what it cannot say
Appendix	Data dictionary and reproducibility

Executive Summary

72% **JOB-TITLE PAY GAP** $4,545 → $7,799	**0.52%** **EDUCATION PAY GAP** PhD vs Bachelor	**0.2%** **GENDER PAY GAP** equity achieved
10.01% **FLAT ATTRITION** 10,010 resigned · CI 9.82–10.20	**3.00 / 2.99** **SATISFACTION** stayed vs resigned — no signal	**r = 0.51** **ONLY REAL SIGNAL** performance ↔ salary

Every company believes it knows its people: “top performers stay because they're engaged”, “we lose people to remote competitors”, “PhDs expect more, so we pay them more”, “tenure earns loyalty”. These stories feel true — but feelings do not survive contact with 100,000 employee records. This report covers ten years, nine departments and seven job titles, and it adds something the narrative version cannot: a SQL layer that recomputes every claim, tests it, and flags the ones that do not hold.

The five things to remember: (1) pay is set almost entirely by job title — a 72% spread — and inside each title the band is a fixed 36.4% of its own minimum, so pay behaves like a formula, not a negotiation; (2) education buys 0.52% and tenure buys nothing; (3) performance ratings are distributed so evenly (~20,000 people in every bucket) that they cannot be measuring performance; (4) attrition is a 10.01% constant — 25 of 26 segments are statistically indistinguishable from the company average; (5) the only “dramatic” trend in the data, a 31% hiring drop in 2024, is not real — 2024 is an eight-month year in this extract.

Why the negative results are the deliverable

A dataset where almost nothing correlates is not a failed analysis — it is a diagnosis: the people processes are running unmeasured. The useful output here is not a prettier dashboard, it is the list of questions this data cannot answer, and the instrumentation needed before the next analysis.

CHAPTER ONE 01

Workforce & Hiring

100,000 employee records across 9 departments and 7 job titles, hired between September 2014 and September 2024. Departments are deliberately even — 10,956 to 11,216 people each, a 2.3% spread — which is the first hint that this extract was assembled rather than observed.

The hiring story: flat, then a false cliff

Headcount arrives at a remarkably steady rate: roughly 834 hires per month for a decade. The naive annual chart shows 2024 collapsing from 9,973 hires to 6,890 — a 31% drop that looks like a restructuring. It is arithmetic, not news: this extract stops on 03 September 2024, so 2024 contains 8 complete months and a three-day stub, while 2014 begins in September. Normalising to complete months, 2024 runs at 851 hires per month against 831 in 2023 — +2.4%, i.e. no cliff at all.

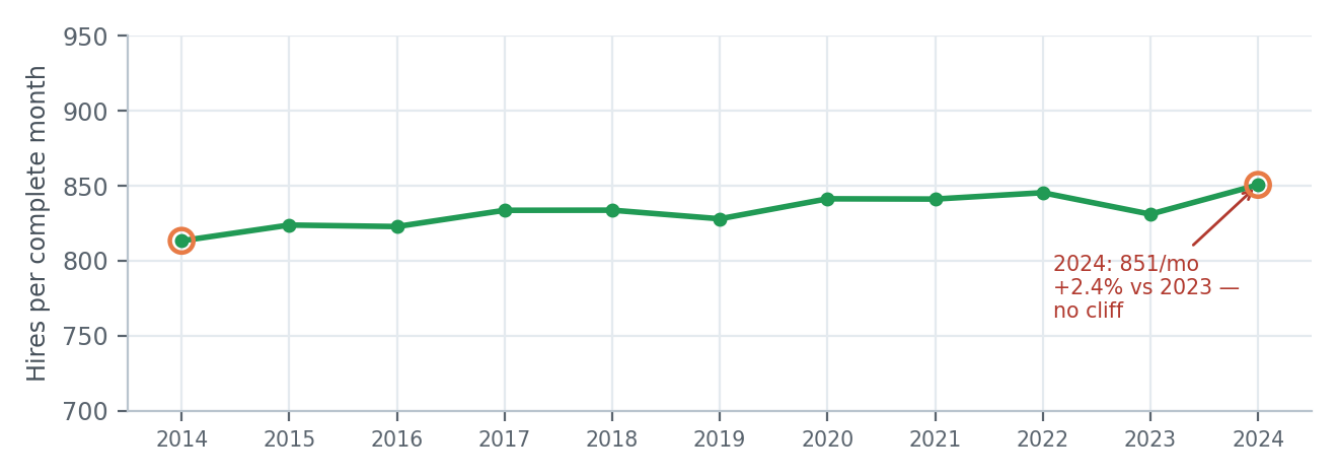


Figure 1 — Hires per complete month. Circled years are boundary years (2014 starts in September, 2024 stops in September) and are compared on rate, never on annual totals.

Correction carried forward from v1

The first edition of this report and README reported “2024 hiring down 31% — a restructuring signal worth a post-mortem” and built a recommendation on it. The drop does not exist. It was a partial-year comparison, and it is now caught by a standing data-quality check (QA6) that every refresh runs before any trend is published.

Who works here

Education level	Headcount	Share	Avg monthly salary	Premium vs High School
PhD	5,051	5.05%	$6,432	+0.50%
Master	14,904	14.90%	$6,417	+0.28%
High School	30,004	30.00%	$6,399	+0.00%
Bachelor	50,041	50.04%	$6,398	-0.01%

Half the company holds a Bachelor's degree and 5% hold a PhD, yet the pay difference between those two groups is 0.52% — about a dollar a day. Education gets people in the door; it does not set the price of the seat.

CHAPTER TWO 02

The Compensation Puzzle

Everyone “knows” the hierarchy of pay: education at the top, then experience, then gender. The data keeps a different scorecard — and only one lever moves.

Job title: the only lever that moves

Engineers and Managers average $7,799; Technicians average $4,545. That is a 72% gap between the top and bottom of seven job titles — larger than every other pay factor in the dataset combined. More revealing is the shape inside each title: the band is always 36.4% of its own minimum (Technicians $3,850–$5,250; Engineers $6,600–$9,000). Pay looks like a title plus a fixed grid, not a market rate negotiated per person.



Figure 2 — Pay range inside each job title. The band is a constant 36.4% of the title's floor in all seven titles: pay is structural, not individual.

Education, gender and tenure: three non-stories

Education is a tie: PhD $6,432 vs Bachelor $6,398 — 0.52%. Gender is the good news: $6,400 (male), $6,405 (female) and $6,415 (other) sit within 0.2% of each other, and after controlling for job title and tenure band (SQL query 09) every gap stays inside ±0.9% — equity is defensible, and worth protecting. Tenure simply does not pay: average salary moves from $6,390 at year 1 to $6,399 at year 9, a 0.5% drift across nine years of service.

> **The one caveat worth stating out loud**
>
> The “year 10 earns the least” line from v1 of this report rested on 44 employees — a rounding-error sample that cannot support a trend. The defensible version is the one above: pay is flat across tenure, which is interesting precisely because it is boring.

Where the money hides: dispersion and a ceiling

5,673 employees — 5.7% of the workforce — sit exactly on the $9,000 monthly ceiling, and the median salary ($6,500) is above the mean ($6,403). Any “top earner” list built on this field is therefore close to arbitrary inside the tie group, which is why the SQL version of that query (Q10) exposes the tie count instead of pretending to rank people.

CHAPTER THREE 03

The Performance Machine That Produces Noise

Ratings on a 1–5 scale are supposed to separate people. Here they separate nobody — and the machinery around them, promotions and training, is just as flat.

A white-picket fence, not a bell curve

Scores 1 through 5 hold 20,120 / 20,013 / 19,999 / 19,940 / 19,928 employees. The difference between the "worst" and "best" bucket is 192 people out of 100,000. Real rating data has a shape — a bulge in the middle, thin tails. This has a pulse.

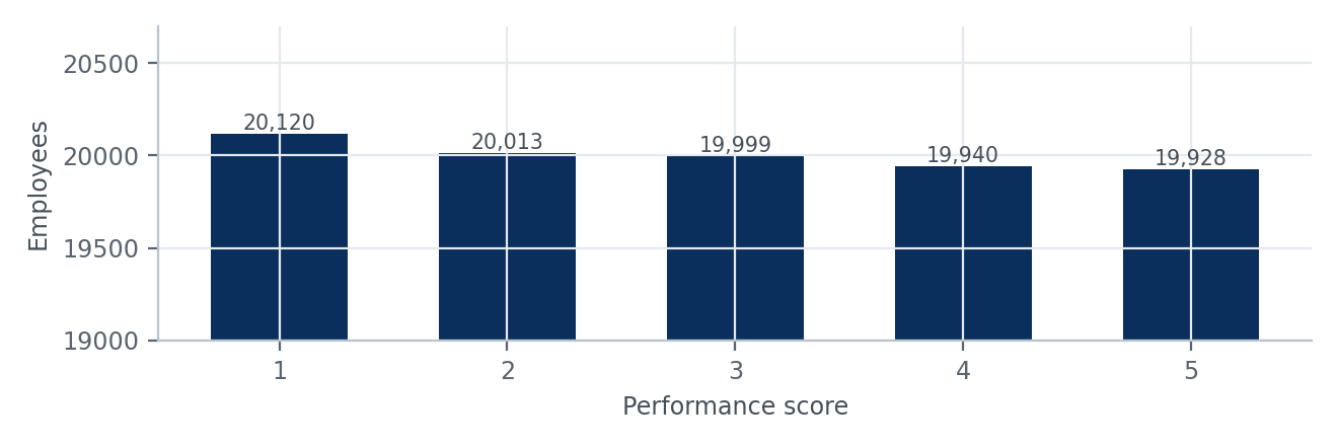


Figure 3 — Performance score distribution: a flat line at ~20,000 per bucket.

Promotions and training: equally flat

Training tier	Headcount	Avg training h	Avg rating	Avg promotions	% promoted	% rated 4+
<20 h	20,053	9.4	3.00	1.00	66.9%	40.0%
20-49 h	30,086	34.7	2.99	1.00	66.4%	39.6%
50-79 h	29,830	64.5	2.99	1.00	66.7%	39.8%
80 h +	20,031	89.5	3.01	1.00	67.0%	40.2%

Every training tier lands on the same 3.0 rating (2.99–3.01) and the same promotion rate (66.4–67.0%), and the company invests a near-identical 49.5 hours per person per year regardless of rating. The only quantity that moves with performance is pay: r = 0.51 between rating and salary, with roughly $493 per rating point in the fitted model. Ratings feed pay; they measure nothing else.

So what is a rating for?

Uniform ratings that still drive pay are worse than no ratings: they convert a measurement problem into a compensation one. The fix is not a new scale, it is evidence anchors, calibration sessions and a forced distribution that survives manager push-back.

CHAPTER FOUR 04

The Attrition Enigma

10,010 people left over the window — 10.01% of the workforce. Finding out who they were is where the story stops being a story.

Department: a one-point spread, and no significance

Finance tops the table at 10.54% and IT bottoms at 9.56% — under one percentage point between the best and worst department, with 95% confidence intervals that overlap the company rate 10.01% everywhere. Read the intervals, not the ranking: with ~11,000 employees per department, a 0.3-point difference is what ordinary sampling noise looks like.

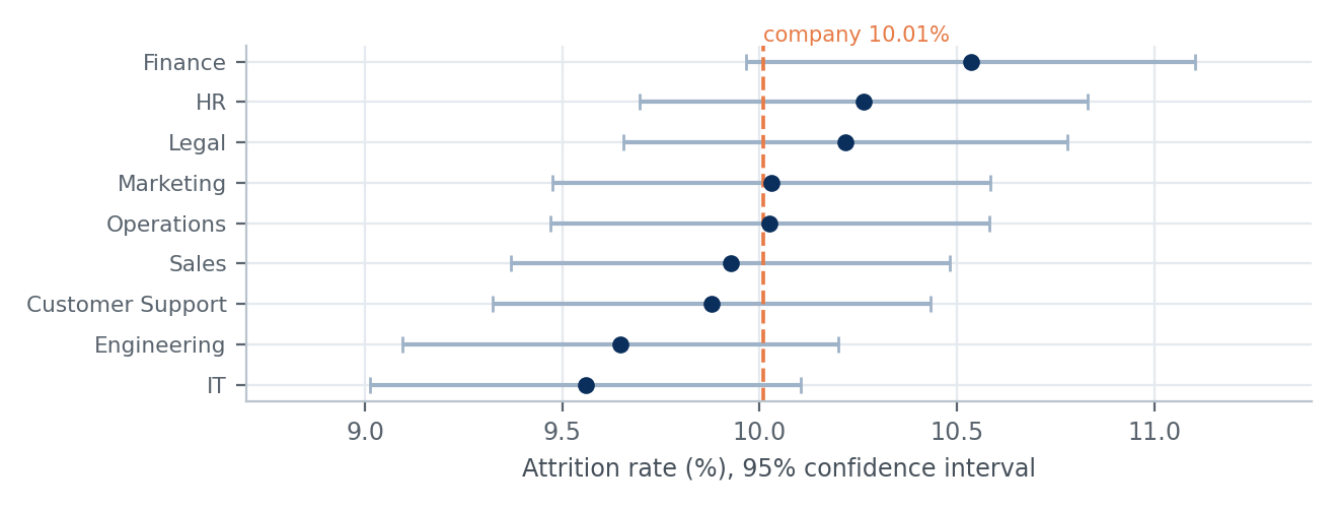


Figure 4 — Department attrition with 95% confidence intervals. Every department's interval crosses the company line (dashed).

Tenure: no early-warning window either

Churn is flat from year 0 (9.79%) to year 9 (9.75%), so there is no “danger zone” to focus retention on. The one cohort that looks different — year 10 at 15.91% — contains 44 employees and is flagged rather than interpreted; it is the group hired in September 2014, and seven resignations out of 44 simply is not a finding.

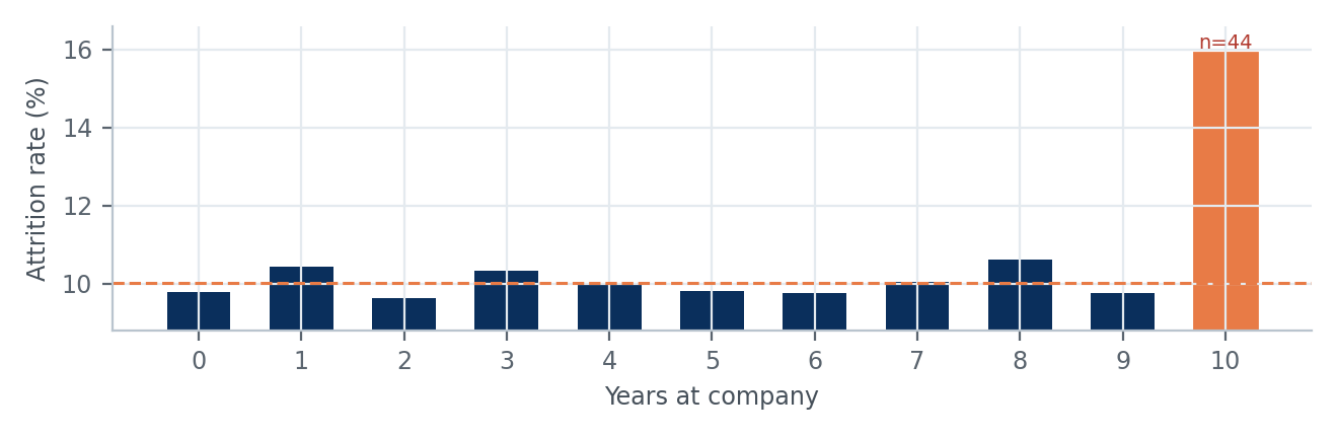


Figure 5 — Attrition by tenure cohort. The flagged cohort (n = 44) is shown for completeness, not for inference.

Who leaves — and who does not

The interesting result is not which segment churns most; it is that no segment does.

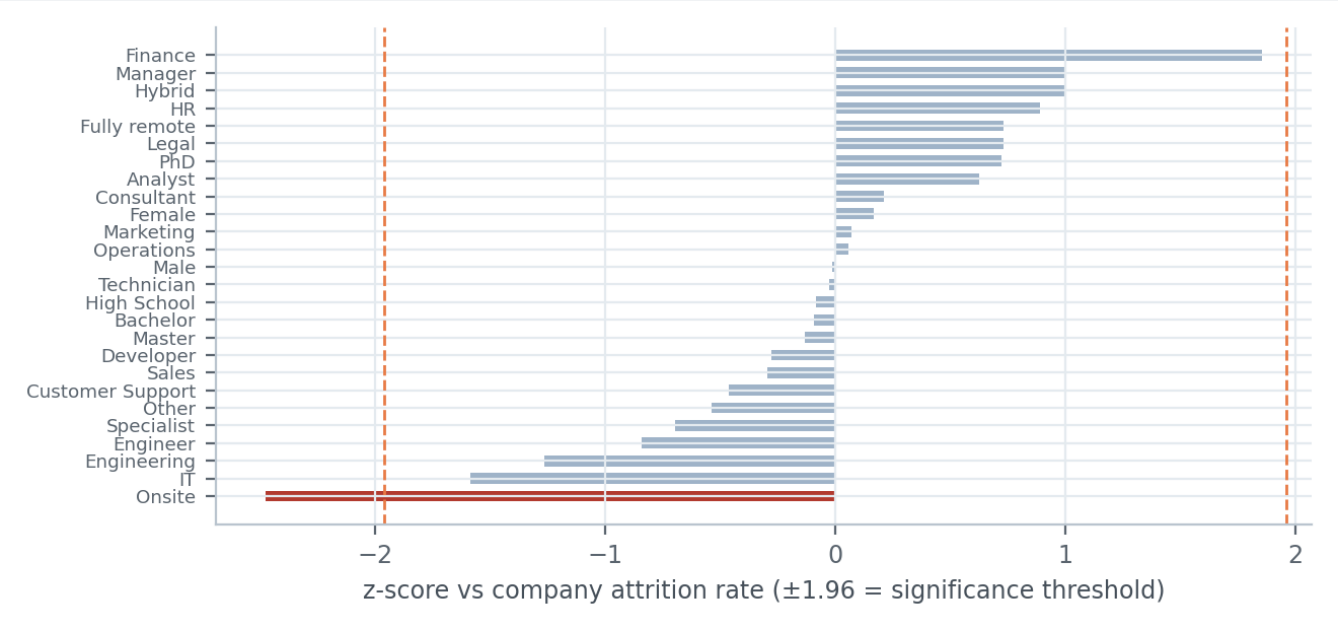


Figure 6 — Significance test of every segment against the company rate. 25 of 26 bars sit inside the ±1.96 band.

All 26 segments, in one line

Across departments, job titles, education levels, genders and work modes, attrition runs from 9.48% to 10.54% — a 1.05 point spread around a 10.01% mean. 25 of 26 segments are statistically indistinguishable from that mean. The single exception is onsite staff (9.48%, z = −2.48): real, but 0.65 points — smaller than a bad month.

Work modes, side by side

Work mode	Headcount	Attrition	z vs company	Avg satisfaction	Verdict
Onsite	19,837	9.48%	-2.48	2.99	significant
Hybrid	60,114	10.13%	+1.00	3.00	no difference
Fully remote	20,049	10.17%	+0.73	2.99	no difference

The satisfaction paradox, and why the model fails

Average satisfaction of employees who stayed is 3.00; of those who resigned, 2.99. On a five-point scale with SD ≈ 1.15, that gap is noise. It is also why both attrition models in chapter 6 plateau at ROC AUC ≈ 0.49 — a coin flip. The exit signal is not weak in this data; it is absent. Either the score measures mood rather than intent, or people disengage quietly while still ticking “3 – satisfied”.

Cost of doing nothing: $23M–$76M a year

About 993 people leave each year at an average annual salary of $76,763. At the standard replacement-cost range of 30–100% of salary that is $23M–$76M annually (estimate — the extract has no replacement-cost field). That is the budget available for retention work, and it is the number worth putting in front of a CFO.

CHAPTER FIVE · NEW

05

The SQL Analytics Layer

Everything above is reproduced in SQL, from a normalised table with constraints, through 24 annotated queries, to a 6-check quality suite — and re-running that pack is what exposed the errors in the first edition of this report.

What was built

- Schema — **employees** with snake_case columns, **employee_id** as primary key (a duplicated extract now fails at load instead of inflating every average), DECIMAL money, CHECK constraints on rating, satisfaction, remote frequency and salary, and a generated **resignation_status** column so “Resigned vs Stayed” has exactly one definition.
- 24 queries in 7 sections — workforce structure, attrition (with confidence intervals and z-tests), compensation and pay equity (controlled for title × tenure band), well-being and work modes, performance/promotion/L&D, NTILE salary quartiles, and trends plus business impact.
- 6 QA checks — row grain and null keys, domain validity, tenure vs hire-date consistency, segment reconciliation (every split sums back to 100,000), salary outlier scan, and snapshot coverage, which is the check that would have caught the phantom 2024 cliff.
- 32 committed output CSVs plus **run_queries.py** — the pack runs head-less in DuckDB (ANSI SQL, no server needed) and rewrites every output; **--check** mode fails if a committed CSV drifts.

What re-running it changed

Query	Bug in the first draft	Reported	Corrected
Q06 / Q07	Attrition divided by company headcount, not the segment's own	Finance 1.18%	10.54%
Q09 / Q11	Pay ratios not multiplied by 100; unguarded division	gap 0.004%	0.4–0.9%
Q10	ROW_NUMBER on a tied salary returned 5,673 “top 3” rows	“top 3”	tie count exposed
Q12	Department average used as an individual benchmark	5,599 underpaid	0 after controlling
Q13	Burnout rate divided by company, not department, headcount	0.38–0.43%	3.4–3.9%
Q17	Filter Promotions >= 0 is always true	100% promoted	66–67%
Q20	Annual hiring compared on a partial 2024	−31% cliff	+2.4% vs 2023

Each correction is annotated in the query file itself and summarised in **SQL/README.md**, because a reviewer's first question about a portfolio number should be “how do I check it?” — and the answer should be a command, not an explanation.

The lesson that transfers to a real job

None of these bugs were visible from the charts. They were visible from arithmetic: segment rates that did not reconcile to the population, a “100%” that could not be true, and an annual comparison across two partial years. Build the reconciliation first; the narrative comes second.

The full query index, design notes and a bug-by-bug corrections log are in **SQL/README.md**; the 32 result CSVs sit in **SQL Outputs/** with the query number in every filename.

CHAPTER SIX 06

What the Models Say

Two models, two very different answers: pay is almost perfectly predictable from the job title, and attrition is not predictable at all. Both results are reported with the metric that can embarrass them.

Can we predict who resigns? No.

Logistic regression (standardised features, C = 0.5) and a random forest (250 trees, depth 8) were both trained on an 80/20 stratified split with a fixed seed. Accuracy looks excellent — 90.0% — until you notice that 90.0% is simply the share of employees who stay: a model that predicts “nobody leaves” scores exactly the same. ROC AUC is the honest metric here, and it lands at 0.490 (logistic) and 0.490 (random forest) — a coin flip. Satisfaction, overtime, training, tenure, team size: none of them carry information about exit.

What sets salary? Essentially one thing.

An ordinary least-squares model on the same feature set reaches $R^2 = 0.9914$ with a mean absolute error of $95 per month. Strip it back to just job title and performance score and it still reaches $R^2 = 0.9914$ — the other twelve columns add nothing. The coefficients read like a price list: Engineer +$2,600, Manager +$2,599, Consultant +$1,951, Developer +$1,300, Specialist +$650, Technician −$649, and +$493 for every extra performance point. Department, gender and education coefficients sit within ±$10 of zero.

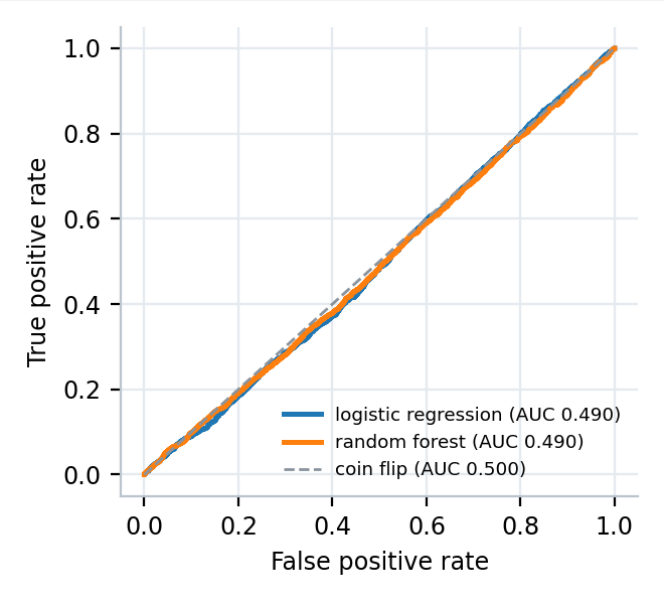


Figure 7 — ROC curves for both attrition models. The dashed diagonal is random guessing; both curves sit on top of it.



Figure 8 — Predicted vs actual monthly salary on the hold-out set. Points collapse onto the 45° line: pay is a formula, not a negotiation.

Model	Task	Primary metric	Value	Reads as
Logistic Regression	Attrition	ROC AUC	0.490	no better than guessing
Random Forest	Attrition	ROC AUC	0.490	no better than guessing
OLS — all features	Salary	R^2	0.9914	explains 99.1% of pay variance
OLS — title + rating	Salary	R^2	0.9914	the other columns add nothing

Why publish a model that fails?

Because the failure is the finding, and because the alternative — reporting the 90% accuracy and calling it a churn predictor — is how HR analytics earns a reputation for noise. Both metrics are shown side by side so the reader can see exactly what the model does and does not know.

CHAPTER SEVEN

07

Myths vs. Math

Five beliefs an HR leader would defend in a meeting, scored against 100,000 records.

HR belief	What 100,000 records show	Verdict	Size
Higher education → higher pay	PhD $6,432 vs Bachelor $6,398	BUSTED	0.52%
Experience → higher pay	Year 1 $6,390 vs Year 9 $6,399	BUSTED	0.5%
Training improves performance	2.99–3.01 avg rating across every training tier	BUSTED	0.02 pts
Unhappy people quit	Stayed 3.00 vs Resigned 2.99	BUSTED	0.01
Remote workers leave more	Onsite 9.48% vs fully remote 10.17%; z = 0.73	MOSTLY	0.7 pt
Ratings differentiate people	19,928–20,120 employees in every score bucket	BUSTED	uniform
Gender pay gap	$6,400 vs $6,405	HELD	0.2%
Job title drives pay	72% spread across seven titles	HELD	72%

Two beliefs survive, and they are the ones worth acting on: role architecture sets pay, and pay equity by gender is genuinely healthy. Everything else in the list is folklore that would survive another decade of meetings — unless it meets a query.

The honest limitation

The distribution of ratings, the perfectly even department sizes and the near-deterministic salary function all point to a synthetic or heavily simulated extract. Read this project as a validated method — schema, queries, significance tests, QA, reporting — demonstrated on data whose real-world conclusions are illustrative.

CHAPTER EIGHT

08

What I Would Tell the CEO

Eight moves, ordered by leverage. Each one is traceable to a query in the SQL pack.

1 — Rebuild job levelling and publish salary bands

The 72% title gap is the entire pay story. Make promotion paths visible — especially for Technicians ($4,545) and Analysts ($5,196), who sit far below the $6,403 average.

2 — Price tenure and skill, not just title

Nine years of service currently buys a 0.5% drift. Time-in-role increments or skill milestones are cheaper than replacing institutional knowledge.

3 — Stop tuning remote policy for retention

Onsite 9.48% · hybrid 10.13% · fully remote 10.17%. The spread is under a point and only onsite clears a significance test — barely. Decide remote policy on hiring reach and cost, not churn.

4 — Rebuild the churn early-warning system

Satisfaction cannot separate leavers from stayers (3.00 vs 2.99) and no field predicts exit (AUC ≈ 0.49). Add intent-to-stay pulses, manager-quality signals and compa-ratio flags, then re-run the model.

5 — Target training instead of rationing it equally

49.5 hours for everyone produces the same 3.0 rating everywhere. Aim hours at low-rated employees and measure lift against a control group.

6 — Calibrate performance reviews

Uniform ratings still drive pay (r = 0.51) while measuring nothing else. Evidence anchors, calibration sessions, and a distribution that survives manager push-back.

7 — Instrument the data before the next analysis

5,673 employees sit on the $9,000 ceiling and title alone explains almost all pay variance. Add exit dates, replacement cost, compa-ratio and engagement-intent fields — otherwise the next analysis will return the same shrug.

8 — Budget retention against the cost of churn

≈993 leavers a year at $76,763 average salary ≈ $23M–$76M in replacement cost (30–100% assumption). That is the fund available for the first seven recommendations.

And protect the wins

Gender pay equity within ±0.2% (and within ±0.9% inside every title × tenure band), near-identical staffing across departments, and a mature tenure mix are genuinely uncommon. Re-audit them quarterly from the SQL pack so they stay wins.

The data didn't give this company more questions. It gave it a to-do list.

CHAPTER NINE 09

Method, Data Quality & Limits

How the numbers were produced, how they were checked, and what they cannot support.

Pipeline

Stage	What happens	Artifact
1 · Load	CSV (100,000 × 20) → normalised MySQL/DuckDB table with constraints and indexes	SQL/01_setup_and_load.sql
2 · Audit	6 QA checks: grain, domains, tenure consistency, reconciliation, outliers, snapshot coverage	SQL/03_qa_checks.sql
3 · Analyse	24 queries: CTEs, window functions, NTILE, z-tests, confidence intervals	SQL/02_analytics_queries.sql
4 · Model	Logistic regression + random forest (attrition), OLS (salary); 80/20 stratified, seed 42	Modelling/model_attrition_and_salary.py
5 · Visualise	21 charts, 4-tab Power BI dashboard, offline Chart.js dashboard	Charts/ · Powe BI Dashboard/
6 · Report	This PDF and its six figures — regenerated from the raw data on every run	Employee Report/build_report.py

Data-quality results

Check	Result
Row grain	100,000 rows, 100,000 distinct employee IDs, 0 null keys
Domain validity	0 violations (score 1–5, satisfaction 1–5, remote ∈ {0,25,50,75,100}, salary > 0)
Tenure consistency	0 mismatches beyond 1 year vs hire_date (592 exact-day rounding cases)
Segment reconciliation	every split sums to 100,000 (difference = 0)
Salary outliers	0 beyond 3σ; 5,673 employees on the $9,000 ceiling
Snapshot coverage	latest year covers 8 complete months → annualise before comparing

What this analysis cannot say

- It is not a time series. Resignation is a static flag with no termination date, so “10.01%” is a decade-wide share of records, not an annual hazard rate.
- It is a snapshot. Hires run from September 2014 to September 2024; the two boundary years are partial and are only ever compared on a per-month basis.
- It is probably synthetic. Even department sizes, a flat rating distribution and salary as a near-deterministic function of title + rating. The methodology is real; the company is not.
- Cost of attrition is an estimate at 30–100% of annual salary, not a booked figure.
- Small cohorts are flagged, not interpreted (tenure year 10, n = 44).

Reproducibility

python3 SQL/run_queries.py rewrites all 32 output CSVs · **python3 "Employee Report/build_report.py"** rebuilds this PDF and its figures · **python3 Modelling/model_attrition_and_salary.py** regenerates every model score and both model figures. No manual steps, no copied numbers.

Appendix · Data dictionary

Source: **Extended_Employee_Performance_and_Productivity_Data.csv** — 100,000 rows × 20 columns, inside Dataset.zip. The SQL table renames these to snake_case and adds a generated status column.

Column	Type	Values / notes
employee_id	INT	Unique employee identifier (primary key)
department	VARCHAR	9 departments, 10,956–11,216 employees each
gender	VARCHAR	Male 48,031 · Female 48,001 · Other 3,968
age	INT	22–60, mean 41
job_title	VARCHAR	7 titles: Engineer, Manager, Consultant, Developer, Specialist, Analyst, Technician
hire_date	DATETIME	2014-09-07 → 2024-09-03 (snapshot)
years_at_company	INT	0–10, mean 4.48
education_level	VARCHAR	High School 30,004 · Bachelor 50,041 · Master 14,904 · PhD 5,051
performance_score	INT	1–5, ~20,000 employees per bucket
monthly_salary	DECIMAL	$3,850–$9,000, mean $6,403, median $6,500
work_hours_per_week	INT	Mean 44.96
projects_handled	INT	0–49
overtime_hours	INT	Mean 14.51
sick_days	INT	0–14, mean ≈ 7
remote_work_frequency	INT	{0, 25, 50, 75, 100} → onsite / hybrid / fully remote
team_size	INT	1–19
training_hours	INT	Mean 49.51
promotions	INT	0–2, mean 1.00 at every rating
employee_satisfaction_score	DECIMAL	1–5, mean 3.00
resigned	BOOLEAN	10,010 true (10.01%)

Files worth opening first

SQL/README.md (design notes + corrections log) · SQL Outputs/23_attrition_segment_significance.csv (the “nothing is different” table) · SQL Outputs/20_hiring_trend_yoy.csv (the corrected trend) · SQL Outputs/06_department_wise_attrition.csv (rates with confidence intervals).